

Chapter 3

The Ability to Elaborate in a Relatively Objective Manner

Introduction

If a person is going to carefully scrutinize the arguments in a persuasive message and thereby follow the central route to persuasion, the person must have the *ability* to evaluate the arguments. Some people may have a greater ability to process a message than others, and some persuasion situations will provide a better opportunity for relatively objective elaboration than others. As we noted in Chapter 2, it is possible to tell whether a target variable enhances or reduces argument processing in a relatively objective manner by manipulating argument quality along with the target variable. If the variable enhances argument processing, subjects' thoughts and attitudes should be more polarized when the variable is present rather than absent, but if the variable reduces argument processing, subjects' thoughts and attitudes should be less polarized when the variable is present rather than absent. In this chapter, we first highlight three variables that appear to affect information processing in a relatively objective way—distraction, message repetition, and recipient posture. Then, we will briefly note some additional variables that may also affect relatively objective information processing.

Factors Affecting the Ability to Elaborate Objectively

Effects of Distraction on Message Elaboration and Persuasion

Research on the effects of distraction on persuasion can be traced to an intriguing experiment by Allyn & Festinger (1961) in which high school students were presented with a speech which argued that teenage drivers are dangerous. The students were either forewarned of the message topic and told that their attitudes would be assessed (attitude orientation) or were simply told that they were to assess the personality of the speaker

(personality orientation). Although these two conditions did not differ in the average attitude change they induced, when analyses were conducted on the most involved subjects (those with extreme attitudes or those who said the issue was important), a significant difference was found such that there was more persuasion in the personality than in the attitude orientation condition. Two possible explanations for this effect were offered. The initial explanation favored by Allyn and Festinger was that the forewarning in the attitude orientation condition stimulated the involved students to counter-argue and/or derogate the source (see also Freedman & Sears, 1965). A second explanation, proposed initially by Festinger and Maccoby (1964) was that the involved subjects in the personality orientation condition were distracted from the counterarguing and/or source derogating that normally would have occurred.

In the 25 years since the Allyn and Festinger experiment, a considerable number of studies have accumulated on both "forewarning" and "distraction," and it is now clear that both effects are viable. In this section we apply the ELM framework to "distraction" and discuss how this variable appears to work by affecting information processing in a relatively *objective* manner. In Chapter 5 we apply the ELM to "forewarning" and address how this variable appears to work by affecting information processing in a relatively *biased* manner.

In 1973, Baron, Baron, and Miller reviewed the accumulated research on "distraction" and concluded that although many individual studies were susceptible to a wide variety of mediational interpretations, there were just two theoretical explanations that could account for the existing data parsimoniously. One explanation was the disruption of counterarguing interpretation favored by Festinger and Maccoby. Another interpretation offered by Baron et al., however, was based, ironically, on Festinger's (1957) theory of cognitive dissonance. Baron et al. argued that distraction manipulations require subjects to exert more effort than usual in order to understand the message. Furthermore, "since choosing to hear a counter-attitudinal message can be viewed as attitude-discrepant behavior, the effort required to comprehend a counterattitudinal message will directly determine the amount of dissonance created by the choice" (p. 317). One way for subjects to reduce this dissonance, of course, is for them to justify their effort by overvaluing the communication.

At the time of the review by Baron et al., the available experiments did not allow a distinction between the two alternative theories because evidence that appeared to support either the counterargument or the dissonance position could also be seen as consistent with the other position. For example, the early Festinger and Maccoby study indicated that distraction enhanced persuasion for counterattitudinal but not proattitudinal messages. Was this because only counterattitudinal messages elicited counterarguments, or because counterattitudinal messages induced more dissonance?

Similarly, the accumulated research suggested that distraction enhanced persuasion only when attention was focused on the message and not on the distraction (e.g., Insko, Turnbull, & Yandell, 1974; Zimbardo, Synder, Thomas, Gold, & Gurwitz, 1970). Was this because message attention enhanced counterarguing, or was it because message attention made the attitude-discrepant behavior of choosing to listen to the advocacy more salient? Importantly, even research using the thought-listing technique, which showed that with increasing distraction the number of counterarguments listed decreased (Keating & Brock, 1974; Osterhouse & Brock, 1970), was open to multiple interpretations. Was a reduction in negative thoughts obtained with distraction because distraction disrupted counterarguing, or was it because distraction induced attitude change via dissonance (or some other process), which was subsequently justified in the thought-listings (Miller & Baron, 1973)?

Testing the Disruption of Message Elaboration Hypothesis

Our initial use of the manipulation of strong and weak arguments (see Chapter 2) came in an experiment that attempted to distinguish the dissonance from the counterargument disruption interpretations of distraction (Petty, Wells, & Brock, 1976, Experiment 1). A second aim of our experiment was to test a more general distraction formulation than "counterargument disruption." Specifically, we reasoned that if the predominant thoughts to a message without distraction were unfavorable, then distraction should disrupt these unfavorable thoughts and lead to increased agreement. However, if the predominant thoughts to a message without distraction were favorable, then distraction should disrupt these favorable thoughts resulting in decreased agreement. Our manipulation of argument quality provides a means of assessing this general "elaboration disruption" hypothesis as well as testing it against the predicted results from dissonance theory.

The disruption of elaboration interpretation holds that distraction should enhance persuasion for a message containing weak arguments (because unfavorable thoughts should dominate under no distraction and would therefore be disrupted), but that distraction should *reduce* persuasion for a message containing strong arguments (because favorable thoughts should dominate under no distraction and would be disrupted). The predictions from dissonance theory are quite different, however. Research on selective exposure and attention indicates that people prefer to hear weak rather than strong arguments against their own position (Kleinhesselink & Edwards, 1975; Lowin, 1967), suggesting that exerting effort to hear strong counterattitudinal arguments induces more dissonance than exerting effort to hear weak ones. Because of this, dissonance theory predicts that for counterattitudinal messages, distraction should enhance persuasion more for strong arguments than for weak ones.

Method. A total of 132 students participated in a 4 (Distraction: no, low, medium, high) $\times$ 2 (Argument quality: weak, strong) $\times$ 2 (Order of attitude assessment: before or after thought-listing) between-subjects factorial design. Subjects were tested in groups while seated in cubicles constructed so that no subject could have visual or auditory contact with other subjects. During any one session, in which one level of distraction was run, some subjects heard strong message arguments while others heard weak message arguments.

Upon arrival at the lab, subjects were told that the experiment concerned the ability to do two things at once. The subjects were told that while they listened to the message, an X would flash periodically in one of the four quadrants of a screen before them. Their task was to record on monitor recording forms the quadrant in which the X appeared. Subjects were either told that "no Xs would flash for now," (no distraction), or the Xs appeared on the screen at 15- (low distraction), 5- (medium distraction), or 3- (high distraction) second intervals during the message. All subjects were instructed to try to pay close attention to the message as they monitored the Xs. It is important to note that the manipulation of distraction was also intended to be a manipulation of effort expenditure (i.e., no, low, medium, and high effort). This was crucial for allowing a test of the dissonance versus disruption of message elaboration interpretations.

Two discrepant messages were prepared for our study. Both messages argued that tuition at the students' university should be increased by 20%, but the messages differed in the presentation of five key arguments. As explained in Chapter 2, the strong arguments were selected so that they elicited primarily favorable thoughts when subjects were instructed to think about them (e.g., raising tuition would allow needed library improvements to be made), and the weak arguments were selected so that they elicited primarily negative thoughts (e.g., raising tuition would allow more trees to be planted on campus). Ratings in a pretest indicated that the messages did not differ in the extent to which they were "difficult to understand," "hard to follow," or possessed "complex structure." Each message was approximately 3 minutes long when recorded. After hearing one of the messages over headphones, subjects completed attitude measures, were given 2.5 minutes to list their thoughts (with the order of attitude and thought assessment counterbalanced), and responded to ancillary questions.

Results. Analyses of manipulation checks revealed that subjects reported feeling increasingly distracted from paying attention to the message and that the task was perceived as increasingly effortful as the number of Xs that flashed increased. Furthermore, analyses revealed that across all levels of distraction, 41% of the thoughts listed in the weak argument conditions were unfavorable, 35% were favorable, and 24% were neutral or irrelevant; whereas in the strong argument conditions 30% were unfavorable, 48% were favorable, and 22% were neutral or irrelevant.

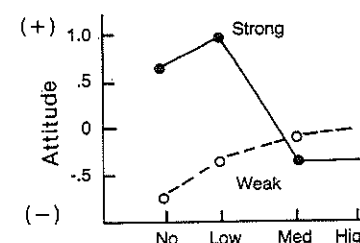


Figure 3-1. Postmessage attitudes as a function of distraction and argument quality (data from Petty, Wells, & Brock, 1976, Experiment 1).

Order of attitude/thought assessment did not affect the attitude or thought measures. The attitude results are presented in Figure 3-1. Consistent with the general elaboration disruption hypothesis, a significant Argument quality $\times$ Distraction interaction was obtained: increasing distraction was associated with more favorable attitudes when the message was weak, but increasing distraction was associated with less favorable attitudes when the message was strong. Importantly, this interaction supports the elaboration disruption hypothesis, but is inconsistent with the effort justification prediction, which expected message acceptance to increase with effort/distraction for both messages.¹

Analyses of the postmessage thoughts listed indicated that overall the messages differed in the number of counterarguments they elicited. In addition, high distraction reduced the production of unfavorable thoughts for the weak, but not the strong message; high distraction tended to reduce the number of favorable thoughts elicited by the strong ($p < 0.10$), but not the weak message. Finally, subjects' recall of the message arguments was assessed. This measure indicated that the recall scores for the no, low, and medium groups did not differ from each other, but the high distraction group recalled fewer arguments than any of the other groups. It is important to note that the attitudinal effects of distraction were most evident between the low and medium distraction groups, consistent with the disruption of message elaboration hypothesis (i.e., the drop in recall did not mediate the effect of distraction on attitudes). Certainly at some high level of distraction, reception of the message could be so severely impaired that this would outweigh other processes (cf., Romer, 1979).

¹In addition, a main effect for message quality was observed (strong arguments induced more agreement than weak arguments). Since this message main effect is also apparent in virtually all of the studies that we report, we will not note this repeatedly for the remaining studies. Instead, we will highlight the more theoretically interesting interaction effects. All ps for effects noted here and throughout the text are < 0.05 unless otherwise noted.

Extending the Generality of the Elaboration-Disruption Hypothesis

The elaboration-disruption hypothesis predicts that, given message reception is comparable across conditions, the dominant thoughts to the message are disrupted by distraction. This effect should hold whether the message topic is proattitudinal or counterattitudinal. Although the results of the previous experiment were consistent with the elaboration-disruption hypothesis, the reduction in favorable thoughts with distraction was only marginally significant. To provide a better test of favorable thought disruption, strong and weak *proattitudinal* messages were employed in a second study (Petty, Wells, & Brock, 1976, Experiment 2).

Method. In this study, 54 students participated in a 2 (Distraction: low or medium) $\times$ 2 (Argument quality: weak or strong) between-subjects factorial design. The procedure and cover story were the same as employed in the preceding experiment, as were the operationalizations of low and medium distraction. These middle levels of distraction were selected because although these levels affected perceptions of distraction and effort, they did not affect message recall. All subjects were exposed to a message arguing that the tuition at their university should be reduced by 50%. As in the first study, two messages were prepared supporting this position, one containing weak arguments and one containing strong arguments. After hearing one of the messages over headphones, all subjects responded to the attitude, thought-listing, and ancillary measures.

Results. Analyses of manipulation checks again revealed that subjects reported feeling increasingly distracted from attending to the message and that the task was increasingly effortful as distraction increased. Furthermore, analyses revealed that consistent with pilot testing, 60% of the thoughts listed in the weak argument conditions were unfavorable, 27% were favorable, and 13% were neutral or irrelevant. In the strong argument conditions 8% of the thoughts listed were unfavorable, 62% were favorable,

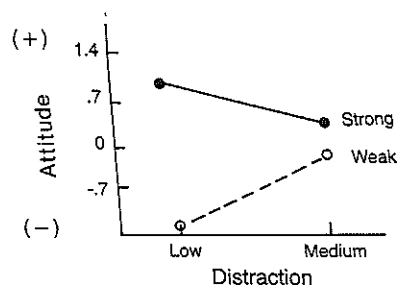


Figure 3-2. Postmessage attitudes as a function of distraction and argument quality (data from Petty, Wells, & Brock, 1976, Experiment 2).

and 30% were neutral or irrelevant. Finally, as anticipated, distraction had no effects on the number of arguments subjects could recall.

The attitude results are presented in Figure 3-2. As in the previous study, the crucial Argument quality $\times$ Distraction interaction was obtained: distraction was associated with increased agreement when the message was weak, but with decreased agreement when the message was strong. Analyses of the postmessage thoughts listed showed that increasing distraction decreased the number of unfavorable thoughts elicited by the weak message and decreased the number of favorable thoughts elicited by the strong message.² As was the case for the preceding study, these data support the elaboration-disruption hypothesis over explanations based on effort justification or message learning.

Additional Support for the Elaboration-Disruption Hypothesis

Two conceptual replications of our distraction studies have been reported. In one, Lammers and Becker (1980) had undergraduates listen to either a counterattitudinal message in favor of doubling tuition (which elicited predominantly unfavorable thoughts) or a proattitudinal message opposed to raising tuition (which elicited predominantly favorable thoughts). After hearing the message under one of the three levels of distraction (judging the pleasantness of irrelevant slides presented at varying rates), subjects rated the perceived extremity of the message's position. Under increasing distraction, subjects tended to see the proattitudinal message as being further from their own position (presumably indicating greater rejection of the message with increasing distraction), and the counterattitudinal message as being closer to their own position (presumably indicating greater acceptance of the message with increasing distraction).

In a closer conceptual replication, Tsai (1984) prepared strong and weak messages (print ads) for a variety of consumer products. As subjects were exposed to the ads via slides, they were either not distracted or were required to count the number of random "clicks" presented on tape. An analysis of subjects' attitudes toward the products revealed a Message quality $\times$ Distraction interaction (see Figure 3-3). Distraction was associated with more favorable attitudes toward the products when the ad arguments were weak, but with less favorable attitudes toward the products when the ad arguments were strong.

The accumulated literature is very consistent with the view that distraction is one variable that affects a person's ability to process a message in a relatively objective manner. Specifically, distraction disrupts the thoughts that would normally be elicited by a message. If the thoughts would normally have been mostly favorable, distraction reduces agreement. If the

²Since the thought data follow and support the attitude data in virtually all of the studies that we report in this volume, we will not detail the thought-listing results for most of the remaining studies. Readers should consult the original reports.

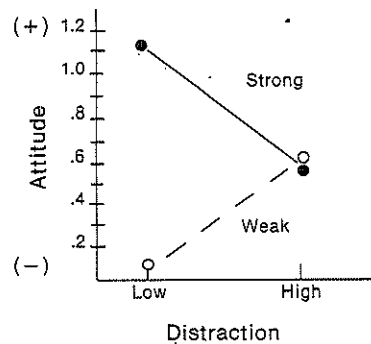


Figure 3-3. Postmessage attitudes as a function of distraction and argument quality (data from Tsai, 1984).

thoughts would normally have been mostly unfavorable, then distraction is associated with enhanced agreement. Distraction should be especially important as a disruptor of message elaboration when people would normally be highly motivated and able to process the message. If motivation and/or ability to process the message would normally be low, distraction should have little effect. The accumulated literature is consistent with this proposition (see Petty & Brock, 1981, for a review).

Effects of Moderate Message Repetition on Message Elaboration and Persuasion

Just as some variables, such as distraction, disrupt message processing, others, such as message repetition, may enhance message processing. Early thinking regarding the effects of message repetition on persuasion came from two sources. One source was from the message learning orientation of Hovland and his colleagues. For instance, Hovland, Janis, and Kelley (1953) stated that "exposure to a persuasive communication which successfully induces the individual to accept a new opinion constitutes a learning experience in which a new verbal habit is acquired" (p. 10). According to Hovland and his associates, then, changing an attitude involves changing a verbal habit. According to this logic, repetition of a persuasive communication, which enhances the likelihood and extent a person attends to, comprehends, and retains the advocacy, should generally increase the likelihood or amount of attitude change, as well (McGuire, 1969).

Evidence consistent with this analysis was provided by McCullough and Ostrom (1974), who presented several different pictorial advertisements to individuals. The presentations of these different advertisements led to more positive evaluations of the product than a single presentation of any one of the advertisements. This study, however, does not address whether the

repeated exposures or the varied advertisements were instrumental in enhancing the regard for the product. Nevertheless, their findings are compatible with Hovland's notion of message learning and attitude change.

A second major framework for understanding repetition stems from the work by Zajonc and his colleagues on mere exposure (see reviews by Harrison, 1977; Sawyer, 1981). Zajonc (1968) reported that a person's attitude toward a stimulus was related positively to its exposure frequency, an effect he attributed to the pleasantness associated with exposure to an increasingly familiar stimulus. Reinterpreting Zajonc's findings, Stang (1975) suggested that these attitudinal effects were mediated by the individual's satisfaction in having learned the attributes of the stimulus (cf., Birnbaum & Mellers, 1979). Although subsequent research by Zajonc and his colleagues (e.g., Kunst-Wilson & Zajonc, 1980; Moreland & Zajonc, 1967) suggests that exposure frequency independently alters both the subjective familiarity and the recognizability of the stimulus, Zajonc (1968) has emphasized that his model applies to nonassociative stimuli (e.g., nonsense syllables, ideographs). More interestingly for theoretical analyses of message repetition is that Sawyer (1981) has argued that Zajonc's mere exposure model may apply to message repetition effects. Also, Stang's (1975) learning perspective on mere exposure, when applied to persuasive communications, is indistinguishable in its predictions from Hovland's message-learning formulation.

Contrary to the predictions of these formulations, however, repetition of meaningful verbal stimuli has not only been shown to increase liking (e.g., McCullough & Ostrom, 1974), but also to decrease liking (e.g., Cantor, 1968; Grush, 1976) and to have no effect on attitudes (e.g., Belch, 1982). Indeed, the most common finding in the persuasion literature is that repeating a persuasive communication tends to first increase and then decrease agreement (e.g., Cacioppo & Petty, 1979b; Calder, & Sternthal, 1980; Gorn & Goldberg, 1980; Miller, 1976).

Based on the ELM, all of these effects are to be expected under prescribable conditions. Specifically, we have proposed that message repetition guides a sequence of reactions to a persuasive communication best conceptualized as a two-stage attitude-modification process (Cacioppo & Petty, 1979b, 1985). In the first stage, repeated presentations of a persuasive communication provide recipients with a greater opportunity to consider the associations and implications of the advocacy in a relatively objective manner. In a persuasion context, subjects should be better able to discern the merits of an advocacy with moderate repetition. Thus, just as distraction can disrupt information processing, moderate message repetition can enhance a person's ability to process the message arguments. The benefit of repetition should be most apparent when additional opportunities are needed to process a message, such as when ability to process the full implications of the message with only one exposure is low (e.g., the message

is not very simple). Once a person has exhausted the associations and has fully considered the implications of the stimulus, the second processing stage becomes dominant. In this stage, attitudinal processes are driven by tedium and/or reactance. Both tedium and reactance will tend to result in decreased message acceptance either by serving as simple negative affective cues or by biasing the nature of message elaboration in a negative direction (see Chapter 5). In this section we explore the consequences of the first (objective) stage of information processing.

It is important to note that in considering the effects of repetition that the number of repetitions required to enhance argument processing but not induce tedium or reactance will depend on a number of factors. For example, the more complex, the more lengthy, or the more rapidly the message is presented, the more repetitions that may be necessary for the full implications of the arguments to be realized. Conversely, the more familiar subjects are with the arguments or information in the message, the fewer the repetitions that may be required to induce tedium or reactance (Batra & Ray, 1986). Thus, what is "moderate" and what is "excessive" repetition will depend on a number of factors (see Cacioppo & Petty, 1985). In our own research, the levels of message repetition appropriate for our specific messages were determined in pilot research.

Testing the Enhancement of Message Elaboration Hypothesis

In order to provide a test of our view that moderate repetition affects persuasion by increasing the opportunity to scrutinize arguments in a relatively objective manner, we conducted a study in which students were exposed to a moderately complex message that was presented either once or three consecutive times (see Cacioppo & Petty, 1985). The message learning, mere exposure, and message elaboration hypotheses can all accommodate repetition increasing message recall. However, the theories differ in their predictions regarding postcommunication attitudes. The elaboration enhancement interpretation predicts the joint outcome that more favorable attitudes are fostered by moderate repetition of strong message arguments, whereas less favorable attitudes are fostered by moderate repetition of weak message arguments. In contrast, the simple message learning hypothesis holds that agreement should increase as learning increases (Stang, 1974); and the mere exposure hypothesis as applied to message repetition (cf., Sawyer, 1981) holds that more positive attitudes develop as the number of exposures increases whether the stimulus is initially evaluated positively or negatively (Zajonc, Shaver, Tavis, & Vankrevel, 1972).

Method. A total of 105 students participated in a 2 (Message repetition: 1 or 3 exposures) $\times$ 2 (Argument quality: strong or weak) between-subjects factorial design. Upon arrival at the lab, subjects were instructed that they were to listen carefully to a message prepared by a campus group using one of several different recording techniques and that their task was to answer questions about the sound quality of the tape following their exposure to the

message. All subjects heard a message advocating that seniors at their university be required to pass a comprehensive exam in their major area as a requirement for graduation. As in our work on distraction, half of the subjects heard a message containing strong arguments and half heard a message containing weak arguments. In addition, half of the subjects heard the message once, and half heard the message three times in succession. Immediately following the presentation of the message, subjects were told that since their attitudes toward the recommendation might affect how they rated the sound quality of the audiotape, we wanted to know how they felt about the issue. Subjects responded to attitude, recall, and ancillary measures.

Results. As expected, subjects recalled more message arguments when the message was presented three times than when the message was presented once. An analysis of subjects' postcommunication attitudes toward the senior comprehensive exam issue revealed an Argument quality $\times$ Repetition interaction (see Figure 3-4) mirroring that found for distraction in the research described above. Specifically, the interaction was a result of the fact that moderate repetition led subjects to express more positive attitudes toward the recommendation when it was supported by a set of strong arguments, whereas moderate repetition caused subjects to express more negative attitudes toward the recommendation when it was supported by a set of weak arguments. This particular interaction pattern is inconsistent with the simple learning or mere exposure accounts of message repetition and supports the view that moderate repetition affects attitudes by enhancing message elaboration.

Extending the Generality of the Elaboration-Enhancement Hypothesis

The studies on distraction and repetition described thus far were characterized by the immediate measurement of attitudes in the same laboratory context in which the persuasive message was presented. Although this procedure allows strict control of irrelevant features and maximizes impact of the independent variables, the effects of elaboration on attitudes should

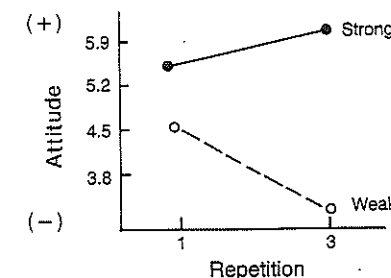


Figure 3-4. Postmessage attitudes as a function of repetition and argument quality (data from Cacioppo & Petty, 1985).

not be confined to this measurement context. In order to extend the generality of our analysis of ability to elaborate, we conducted a second study in which the crucial attitude measure was dissociated in time and setting from the message presentation (Cacioppo & Petty, 1980a, Experiment 2).

Method. A total of 89 students participated in a 3 (Message repetition: 1, 3, or 5 presentations) $\times$ 2 (Argument quality: strong, weak) between-subjects factorial design.³ In this study, students were exposed to a strong or weak message in favor of raising the price of their local newspaper. The message was presented to subjects as an audiotape of a telephone interview with a local resident. The strong message emphasized the benefits subscribers would receive from the price increase, whereas the weak message emphasized the benefits to management. The message was played either 1, 3, or 5 times in succession, and as in the previous study, subjects were instructed to evaluate the sound quality of the tapes.

Immediately following exposure, subjects listed their thoughts about the tapes, rated the sound quality, and recalled all of the message arguments they could remember. From 8 to 14 days later, individuals were contacted by a phone interviewer who appeared unrelated to the initial experimenter. The second experimenter, who was blind to the respondents' initial experimental conditions, inquired about a number of community issues including attitudes toward increasing the price of the local paper.

Results. Consistent with the previous study on message repetition, recall of both strong and weak message arguments increased with repeated exposures. Analyses of the thought-listings indicated that the strong arguments elicited more favorable thoughts and nonsignificantly fewer unfavorable thoughts than the weak arguments. Importantly, analyses of the attitude data revealed an Argument quality $\times$ Repetition interaction (see Figure 3-5). Consistent with the elaboration-enhancement hypothesis, the interaction was a result of the fact that moderate repetition led subjects to express more positive attitudes toward the recommendation when it was supported by a set of strong arguments, whereas moderate repetition caused subjects to express more negative attitudes toward the recommendation when it was supported by a set of weak arguments.

Additional Support for the Elaboration-Enhancement Hypothesis

Corlett (1984) recently reported a conceptual replication of our repetition studies using mock television advertisements rather than audiotaped editorials. Corlett developed strong and weak versions of advertisements

³For exploratory purposes, a third group of subjects received a message containing novel arguments that were weak but "subtly contradictory." Subjects exposed to this message showed an inverted-U pattern of acceptance with repetition.

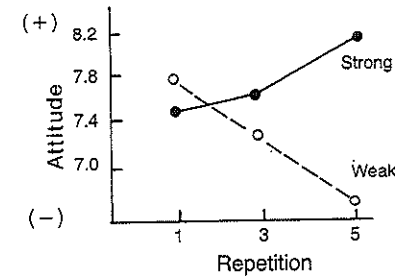


Figure 3-5. Delayed attitudes as a function of repetition and argument quality (data from Cacioppo & Petty, 1980b).

promoting a little-known beer. Half of the subjects were exposed to the strong version of the advertisement embedded among other commercials in a 60-min television show, and half of the subjects were exposed to the weak version of the advertisement. Results indicated that male undergraduates expressed more positive attitudes toward the beer as exposures increased from 1 to 3 when the ad-arguments were strong, but they expressed more negative attitudes as repetition increased when the ad-arguments were weak (see Figure 3-6).⁴ Interestingly, repetition had no significant impact on the attitudes of female students. These results are interesting in light of the fact that males are the primary consumers of beer and, therefore are more likely to view advertisements regarding beer as personally relevant. As we have already noted and will document in the next chapter, both motivation (e.g., personal relevance) and ability (e.g., distraction, repetition) variables are important determinants of the extent of message elaboration.

Effects of Recipient Posture on Message Elaboration and Persuasion

Although evidence continues to accumulate regarding how nonverbal bodily cues emanating from a communicator can affect perceptions of the source (e.g., Argyle & Kendon, 1967; Zuckerman, DePaulo & Rosenthal, 1981; Mehrabian, 1981) and thereby persuasion (e.g., McGinley, LeFevre, & McGinley, 1975), little attention has been paid to how the bodily orientation of a message recipient affects message processing and persuasion. In an exploratory study we had undergraduates listen to a cogent message advocating a 20% increase in tuition at their university while they were positioned in one of several postures—standing, sitting, or reclining (Petty,

⁴When repetitions were increased to more extreme levels (5 and 7 exposures), subjects hearing the strong arguments showed a slight (nonsignificant) decline in attitudes, and subjects exposed to the weak arguments showed a more substantial decline. See Chapter 5 for discussion of excessive repetition.

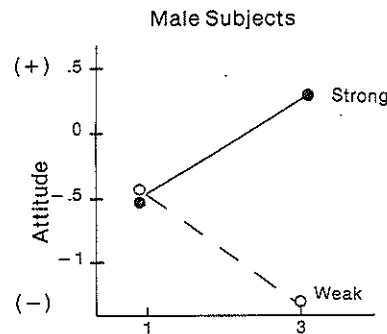


Figure 3-6. Postmessage attitudes for male subjects as a function of moderate repetition and argument quality (data from Corlett, 1984).

Wells, Heesacker, Brock, & Cacioppo, 1983, Experiment 1). The results of this study indicated that subjects who were reclining on a cushioned table while listening to the message expressed more agreement with a tuition increase than subjects who were standing during message exposure. Interestingly, although subjects who were seated during message exposure indicated that they were the most "comfortable," they were no more persuaded than standing or supine subjects. Additionally, the insignificant correlation between rated comfort and message agreement ($r = 0.07$) provided evidence against the view that posture influenced attitudes simply by inducing a pleasant affect or mood that became associated with the advocacy. It was possible, however, that the supine posture was associated with greater message elaboration. Since the arguments in the message were strong, the results of greater scrutiny would be enhanced agreement.

Testing the Message Processing Hypothesis

In order to provide a specific test of the elaboration interpretation, we conducted a second study in which recipient posture was manipulated along with the quality of the issue-relevant arguments contained in the message (Petty, Wells, Heesacker, Brock, & Cacioppo, 1983, Experiment 2).

Method. In this study, 65 female students participated in a 2 (Body posture: standing or reclining) $\times$ 2 (Argument quality: strong or weak) between-subjects factorial design. Subjects were told that they were participating in a cooperative investigation between the psychology department and Tech-Headphones, Inc. It was explained that the headphone company was funding a study that would provide information about basic auditory processes and about how to design headphones to provide maximum listening pleasure in a wide variety of naturalistic settings. Following these instructions, subjects were led to individual cubicles where they were

instructed either to stand or to lie down on a cot while listening through their headphones. The subjects were told that they would hear a brief excerpt from a recent campus radio broadcast. After they had assumed the appropriate posture, they heard several minutes of instrumental music followed by an editorial message arguing that seniors should be required to pass a comprehensive exam in their major area as a requirement for graduation. After hearing either the strong or the weak arguments version of this editorial, subjects expressed their attitudes on the issue, listed their thoughts, rated the headphones, and responded to ancillary measures.

Results. Subjects reported that they were more comfortable in the reclining than in the standing posture, and that they were less "distracted" from thinking about the editorial in the supine position. The attitude results are presented in Figure 3-7. Of greatest interest was the appearance of a significant Argument quality $\times$ Posture interaction that provided support for the view that a reclining posture facilitates (or a standing posture disrupts) careful processing of a message. A decomposition of the interaction revealed that subjects did not attitudinally distinguish the strong from the weak arguments when they were processed in the standing position, but that the arguments did differ significantly in their persuasibility when processed in the reclining posture.

Unanswered Questions about Posture

Although it appears clear that posture is capable of influencing message elaboration and thereby persuasion, at least two important questions about recipient posture remain unanswered. One concerns *why* a reclining posture facilitates and/or a standing posture inhibits message-relevant thinking. One possibility is that reclining leads to a reduced degree of self-focus compared with standing (Wegner & Giuliano, 1980), and the less attention allocated to the self, the more that is available for processing information from the external environment (Scheier, Carver, & Matthews, 1983).

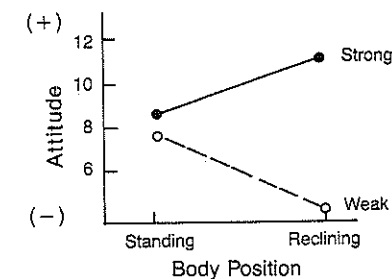


Figure 3-7. Postmessage attitudes as a function of recipient posture and argument quality (data from Petty, Wells, Heesacker, Brock, & Cacioppo, 1983, Experiment 2).

Another possibility is that subjects are more willing to undertake cognitive work when their mood is positive (reclining) rather than negative (standing). A second important question concerns the limitations on the posture effect. In our research, the message recipients were in the reclining posture for only a brief period of time. If they had remained in the supine position for a relatively long period of time, perhaps relaxation would have become so great that processing would have been inhibited. Although a wide variety of interesting questions remain to be addressed concerning recipient posture and persuasion, it does appear that under some circumstances at least, body posture can affect the extent of message processing and thereby persuasion. We briefly consider the posture effect again in Chapter 8.

Other Variables Affecting Ability to Process Objectively

The variables of distraction, repetition, and body posture were considered to affect information processing in a relatively objective manner because they tended to disrupt or facilitate whatever the dominant cognitive response to the message was. When the arguments were strong, favorable thoughts were disrupted or facilitated, but when the arguments were weak, unfavorable thoughts were disrupted or facilitated. Other variables studied by persuasion researchers may also be important because of their effects on a person's ability to elaborate a message. We note a few of these below.

* *Message Complexity/Comprehensibility*

Researchers have studied the complexity or comprehensibility of message arguments (e.g., Eagly, 1974; Regan & Cheng, 1973) mostly in the context of McGuire's (1968, 1969) information-processing model. The ELM holds that the strengths of cogent arguments and the flaws in specious ones should become more apparent as complexity is reduced and comprehensibility is increased. Thus, these variables, like repetition, distraction, and posture should be able to enhance or reduce agreement depending upon whether the message arguments are strong or weak. Importantly, our analysis assumes that subjects are also motivated to process the persuasive message (see Chapter 4). If motivation was low, then message factors such as complexity and comprehensibility might affect agreement by serving as simple validity cues via the peripheral route. For example, consider a person who is unmotivated to process a message that employs difficult vocabulary. To the extent that the message recipient takes the complex vocabulary as a sign of the author's intelligence and credibility, agreement might be increased as a result of this complexity cue.

Message Modality

In addition to message factors that are capable of affecting the ability to process such as complexity, the modality of message presentation may also be important. For example, research indicates that messages presented on audio or videotape may be more difficult to process than the same message

presented in print (e.g., Chaiken & Eagly, 1976; see Wright, 1981). In print, the recipient may process the message at an optimal pace, stopping to consider difficult points and elaborating at will. On radio or television, however, the pace of the message is externally controlled. Unless the message is very simple, complete elaboration may not be permitted with one exposure. Based on this analysis, the ELM would predict that given sufficient motivation to process an advocacy, print should generally enhance agreement to strong but reduce agreement to weak messages compared with the same message presented on tape.

Heart Rate

Cacioppo, Sandman, and Walker (1978) proposed that transient and specific accelerations in a message recipient's heart rate could facilitate the processing of a communication. This analysis was based on the accumulated evidence that phasic heart rate deceleration is associated with increased sensory sensitivity, and phasic heart rate acceleration is associated with greater cognitive work (e.g., Lacey & Lacey, 1974; Lacey, Kagan, Lacey, & Moss, 1963). In an initial test of this idea, Cacioppo, Sandman, and Walker (1978) operantly conditioned subjects to momentarily accelerate and decelerate their heart rates without altering respiratory or somatic activity. After successful training, subjects were presented with brief counterarguable messages during periods of acceleration, deceleration, and basal heart rate. The major result of this study was that cardiac acceleration was associated with the increased production of counterarguments and reduced agreement with the advocacy.

In a more definitive test of the hypothesis that transient and specific heart rate could affect message processing, Cacioppo (1979) conducted an additional persuasion study in a cardiology clinic using fully informed outpatients who had implanted cardiac pacemakers. The patients' pacemakers were "demand type," so called because they paced the heart at a constant rate (72 bpm) when natural pacing produced a rate below the set level. However, the pacemaker level could be accelerated to 88 bpm when an uncapped magnet was placed appropriately over the pacemaker. In this study, senior citizens read counterarguable communications on two involving issues under basal and accelerated heart rate conditions although subjects were unaware of when this manipulation occurred. The major result of this study was that subjects generated significantly more counterarguments when the message was read under accelerated than under basal heart rate conditions. In both studies of cardiac activity, then, the persuasive messages employed elicited primarily unfavorable thoughts, and accelerated heart rate was associated with enhanced negative thinking. The ELM predicts that if transient and specific heart rate acceleration enhances relatively objective processing, then enhanced heart rate should be associated with the production of more unfavorable thoughts if the message arguments are counterarguable and weak, but more favorable thoughts if

the message arguments are strong. The latter possibility has not yet been examined.

Recipient Intelligence/Education

Two potentially important individual difference variables that have received attention over the years are the general intelligence and educational levels of a message recipient (e.g., Eagly & Warren, 1976; Hovland, Lumsdaine, & Sheffield, 1949; Janis & Rife, 1959; see review by McGuire, 1969). Based on the ELM, one might be tempted to propose that the greater a person's general intelligence or education, the greater the person's ability to process a message. However, would this processing be relatively objective or relatively biased? If a message contained strong or weak arguments that could be judged on the basis of logic alone, then intelligence or training in logic or reasoning would be expected to show the pattern indicative of objective processing—especially if the chain of reasoning required to evaluate the message was relatively complex (i.e., intelligence or education would be associated with more agreement to strong and less agreement to weak arguments). However, as we noted in the previous chapters, perceptions of argument strength are often quite subjective and idiosyncratic. The same argument may be seen as strong or weak depending upon the store of prior information available in memory. To the extent that intelligence or education is associated with more extensive attitude-congruent knowledge, this could give people greater ability to process arguments in an attitude-consistent fashion (see Chapter 5). Intelligent or highly educated people may also be more motivated to defend their beliefs because perceived correctness may be an integral part of their self-esteem (cf., McGuire, 1968).

Finally, it is interesting to note that intelligent or well-educated people may be more susceptible to particular peripheral cues than people of less intelligence or education in certain situations. For example, intelligent or educated people (because of their inherent reasoning powers, knowledge, or training) may be more likely to believe that there are two reasonable sides to most issues. Because of this, when an issue arises for which motivation or ability (e.g., prior knowledge) are low, intelligence or education may be associated with greater reliance on the cue value of a two-sided message. Specifically, these subjects may be more likely to reason that a source that is aware of both sides is more credible (intelligent or knowledgeable) than a source that presents only one side. On the other hand, people of less intelligence or education may be more likely to reason that the source is confused. Because of the sparse literature on intelligence and educational level, we will not discuss these variables further (except to distinguish intelligence from need for cognition in Chapter 4). However, in Chapter 5 we discuss the impact of a variable that is closely related—the extent of prior knowledge on an issue.

Retrospective

In this chapter we reviewed evidence consistent with the proposition that some variables affect persuasion by modifying a person's ability to elaborate issue-relevant arguments in a relatively objective manner. Thus, when external distraction accompanies a communication and this distraction engages cognitive capacity, people become less able to elaborate the merits of the arguments presented. The net result is that strong arguments are not accepted as much as they would be without distraction, but weak arguments are not rejected as much as they would be without distraction. Likewise, strong arguments are accepted more when they are repeated a moderate number of times, but weak arguments are rejected more when repetition allows full elaboration of their flaws. Among the other variables that may affect a person's ability to process a message in a relatively objective manner are message complexity/comprehensibility, the modality of message presentation, and recipient factors such as body posture and phasic heart rate accelerations. In the next chapter we discuss variables that modify a person's *motivation* to process a message in a relatively objective manner.